Syllabus for the Exam of JOA(IT) to be conducted by HPRCA

A. SUBJECTS AS PER ESSENTIAL QUALIFICATION

a) COMPUTER

I. FUNDAMENTALS OF COMPUTER:

- Definition & Characteristics of Computer
- History & Generation of Computers
- Types of Computers (Analog, Digital, Hybrid)
- Classification by Size (Micro, Mini, Mainframe, Super)
- Basic Computer Organization
- Block Diagram of Computer :Input → Process → Output concept
- Applications of Computer in Daily Life

II. COMPUTER HARDWARE:

- Input Devices — Keyboard, Mouse, Scanner, Microphone, Webcam, Light Pen.
- Output Devices — Monitor, Printer, Speaker, Projector.
- Types of Printers (Inkjet, Laser, Dot Matrix)
- System Unit — CPU, Motherboard
- CPU Components — ALU, CU, Registers
- Primary Memory — RAM & ROM (Types)
- Secondary Storage — HDD, SSD, Pen Drive, CD/DVD
- Memory Units — Bit, Byte, KB, MB, GB, TB

III. COMPUTER SOFTWARE:

- Definition & Need of Software
- Types of Software — System Software (OS, Device Drivers, Utilities)
- Application Software (Word Processor, Spreadsheet, etc.)
- Programming Language Software
- High Level Language, Assembly Level Language, Machine Level Language
- Difference between Hardware and Software Freeware, Shareware, Open Source Software
- Software Installation & Licensing Basics

IV. OPERATING SYSTEM:

- Definition & Functions of Operating System
- Types of OS — Single User, Multi User, Multitasking, Time Sharing
- Popular Operating Systems — Windows, Linux, MacOS, Android
- Windows Basics: Desktop, Taskbar, Start Menu, Icons
- File & Folder Management, Control Panel Settings
- Recycle Bin, Shortcut Keys
- Booting Process (Cold Boot & Warm Boot)
- Basic DOS Commands (Internal & External)

V. MICROSOFT OFFICE

a. MS Word:

- ✓ Features & application of Word Processing , Creating, Saving & Opening Documents
- ✓ Formatting Text , Short Cut Keys, Hot Keys, Find and Replace
- ✓ Paragraph Formatting (Alignment, Spacing, Indent)
- ✓ Page Setup (Margins, Orientation, Page Size)
- ✓ Header & Footer, File Protection, Creating Lists, Numbering

- ✓ Tables — Insert, Merge, Split
- ✓ Spelling & Grammar Check, Thesaurus, Mail Merge Basics, Macro
- ✓ Print Preview & Printing

b. MS Excel

- ✓ Features and benefits of Spreadsheet , Workbook & Worksheet Concepts
- ✓ Entering & Editing Data, Cell/Row/Column Operations
- ✓ Functions: Mathematical, Statistical, Logical & String Function
- ✓ Cell Formatting & Borders
- ✓ Charts & Graphs, Pivot Table, Goal Seek, Scenario
- ✓ Sorting & Filtering Data, Print Area & Page Setup

c. MS PowerPoint

- ✓ Creating & Saving Presentation, Adding Graphics, Multimedia
- ✓ Slide Layout & Design Themes
- ✓ Inserting Text, Images, Tables
- ✓ Slide Transitions & Animations Effects
- ✓ Slide Master, Hyperlinks and Action Button
- ✓ Slide Show View, Printing Slides & Handouts

VI. INTERNET & NETWORKING:

- ➢ Definition & Uses of Internet, History of Internet
- ➢ WWW, Website, Web Page, Web Browser
- ➢ Popular Browsers (Chrome, Firefox, Edge)
- ➢ URL, HTTP, HTTPS, IP Address, Domain Name
- ➢ Search Engines ,Emails,Social Media Basics
- ➢ Types of Networks — LAN, MAN, WAN
- ➢ Network Devices — Router, Modem, Switch, Hub
- ➢ Wired vs Wireless Networks, ISP (Internet Service Provider)

VII. CYBER SECURITY & DIGITAL LITERACY:

- ➢ Computer Virus, Worm, Trojan Horse
- ➢ Antivirus Software & Firewall
- ➢ Types of Cyber Crimes
- ➢ Safe Internet Browsing Practices
- ➢ Password Security & Data Protection
- ➢ Phishing & Spam Awareness
- ➢ Digital Payments Basics (UPI, Net Banking, e-Wallet)
- ➢ e-Governance & Government Online Services
- ➢ Cloud Computing (Basic Concept)
- ➢ Digital Footprints, Intellectual Property Rights

VIII. NUMBER SYSTEM:

- ➢ Different Number System
- ➢ The Decimal Number system
- ➢ The Binary Number System
- ➢ The Octal Number System
- ➢ The Hexadecimal Number System
- ➢ All Conversion between Number System
- ➢ ASCII Code
- ➢ Basic Logic Gates

IX. DBMS: Introduction to Database:

- ➢ DBMS purpose and advantages of DBMS, disadvantages
- ➢ Comparison between traditional file systems and DBMS.
- ➢ Data independence (logical and physical)
- ➢ Three-level architecture of DBMS.

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- Entity, Tuple, Attribute, Domain, Degree, Cardinality, and different types of keys.
- Overview of MySQL, features of MySQL, uses of MySQL, Creating database, selecting database, deleting database, showing databases, data types, constraints.
- SELECT statement, WHERE clause, relational operators, logical operators, sorting data (ORDER BY), removing duplicate records (DISTINCT).

X. WEB Technologies:

a. HTML Introduction:

- ✓ Structure of HTML Program, Heading Styles, Text Styles,
- ✓ Lists : ordered and Unordered Lists
- ✓ Adding Images, Creating Tables
- ✓ Hyperlinks (internal & external)
- ✓ Form tags and its elements

b. Introduction to PHP:

- ✓ Important tools and software requirements (like Web Server, Database, Editors etc.)
- ✓ Scope of PHP, basic syntax, PHP variables and constants, data types , Expressions, PHP operator precedence and associativity.
- ✓ Control statements (conditional statements, switch case, loops, break & continue)
- ✓ Handling HTML form using various form control with PHP,
- ✓ PHP Functions: Function, Need of Function, declaration and calling of a function, call by value & call by reference.
- ✓ Arrays, Strings, Regular Expression.

b) MATHEMATICS:

Number System, Profit & Loss, Simplification, Simple & Compound Interest, Percentage, Time & Work, Ratio & Proportion, Time-Speed & Distance, Average, Data Interpretation, Algebra, Mensuration.

2. General Awareness

i. **General knowledge**: General Knowledge including General knowledge of Himachal Pradesh.

ii. **Current Affairs** .

iii. **Everyday Science** .

iv. **Logical Reasoning** .

v. **ENGLISH :** Reading Comprehension Word Power, The Sentence, Subject And Predicates , Articles, Number, Gender, Punctuation, Comprehension, Noun, Pronoun, Verb, Finite and Non-finite Verb, Adverb, Adjective, Preposition, Conjunction, Interjection , Error Correction, Sentence Rearrangement, Vocabulary, Antonyms, Synonyms, Tenses, Subject-Verb Agreement, Idioms, Modals, Active Voice & Passive Voice, Change the narration: Direct and Indirect, Phrases and a Clauses , One word substitution, Transformation of Sentences.

vi. **हिन्दी** भाषा–व्याकरण एवं लिपि का परिचय; वर्ण विचार एवं आक्षरिक खण्ड; शब्द विचार

(क) परिभाषा के आधार पर (तत्सम्, तद्भव, देशज, विदेशी), रचना के आधार पर, प्रयोग के आधार पर, अर्थ के आधार पर।

(ख) विकारी– संज्ञा, सर्वनाम, क्रिया, विशेषण, अविकारी– क्रिया– विशेषण, सम्बन्ध बोधक, समुच्य बोधक, विस्मयादि बोधक ; पद परिचय ; शब्द शक्तियां ; शब्द रूपान्तर– लिंग, वचन, कारक, काल, वाच्य ; संधि; समास; उपसर्ग; प्रत्यय ; वाक्य विचार, अर्थ विचार (पर्याय, विलोम, वाक्यांश के लिए एक शब्द, समानार्थी); विराम चिन्ह; शुद्धिकरण (शब्द शुद्धि, वाक्य शुद्धि) ; मुहावरे एवं लोकोक्तियां; अलंकार।

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